CHEM 427/527 · SESSION 9 · THURSDAY, OCTOBER 29, 2026

Experimental design and statistical inference

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Compute the 90th percentile of first-draw lead by the Lead and Copper Rule’s rank method, check it from the sorted values, and say why the rule uses it rather than the mean.
2. Put a one-sided 95% lower bound on the 90th percentile, and say whether it shows an exceedance of 15 and of 10 µg/L.
3. Choose a paired or a two-sample t-test for first-draw against flushed lead from how the samples were taken.
4. Say whether a significant difference is a meaningful one, in µg/L against the action level.
5. Ask an assistant for a value and for a printout that would show it wrong, decide first what you will accept, and read the value against it.

FIRST, EVERYONE PULLS

Get session 9 with git pull

`git pull`

S09-inference appears, with

- **data/fetch_flint.py**: downloads the lead results and checks them
- **data/draws.csv**: 36 draws of 60 samples; each of you gets one
- **data/README.md**: where the data come from

Session 8 is due today

- Upload it by 1:30 pm.
- Fill in your prompt record, then Restart and Run All before you upload.

The lead results are not in the repository: the first code cell of section 1 downloads them.

FROM ONE HOME TO ONE CITY

What one number decides whether a city is over the action level?

Session 8

A lab reported each home’s lead as a number, or as zero below its detection limit.

Today

A regulator holds 271 of those numbers from one city.

The question

Which value, set against the action level, decides whether the city is over?

The mean, the median, the highest home, or something else?

THE DECISION

Is the city over the lead action level?

The rule

40 CFR 141.80(c): a 90th percentile above the action level is an exceedance. 15 µg/L in 2015; 10 µg/L from November 2027.

Wrong one way

Residents keep drinking leaded water, which is what happened in Flint in 2015.

Wrong the other

A city replaces lines on the noise of which homes were sampled.

Real: Virginia Tech’s Flint Water Study, 2015

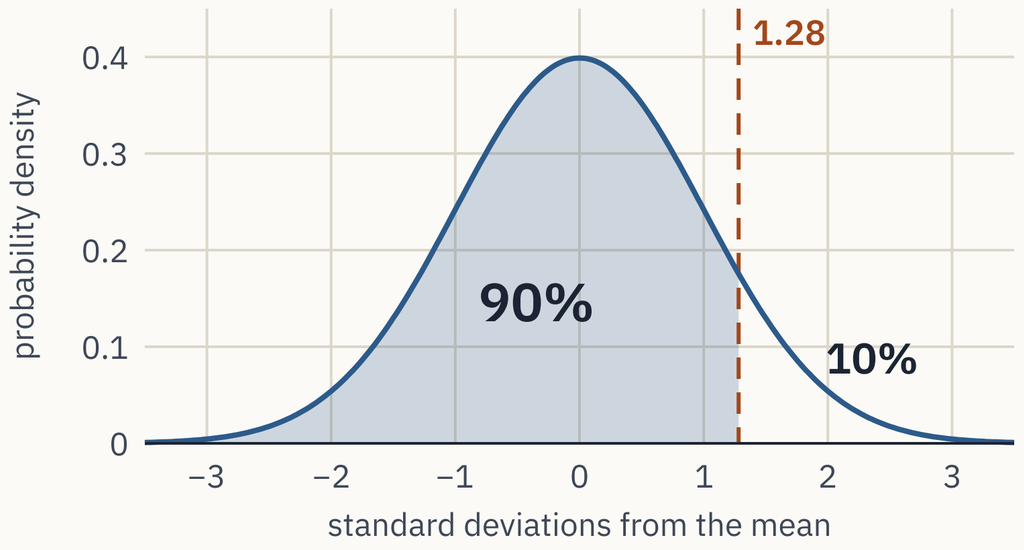
271 tap-water samples from 269 homes, August and September 2015, lead by ICP-MS: a first draw after at least 6 hours of stagnation, then samples after 45 s and 2 min of flushing. Unaltered, downloaded from the study’s site.

The action level is fixed before anyone looks at the data.

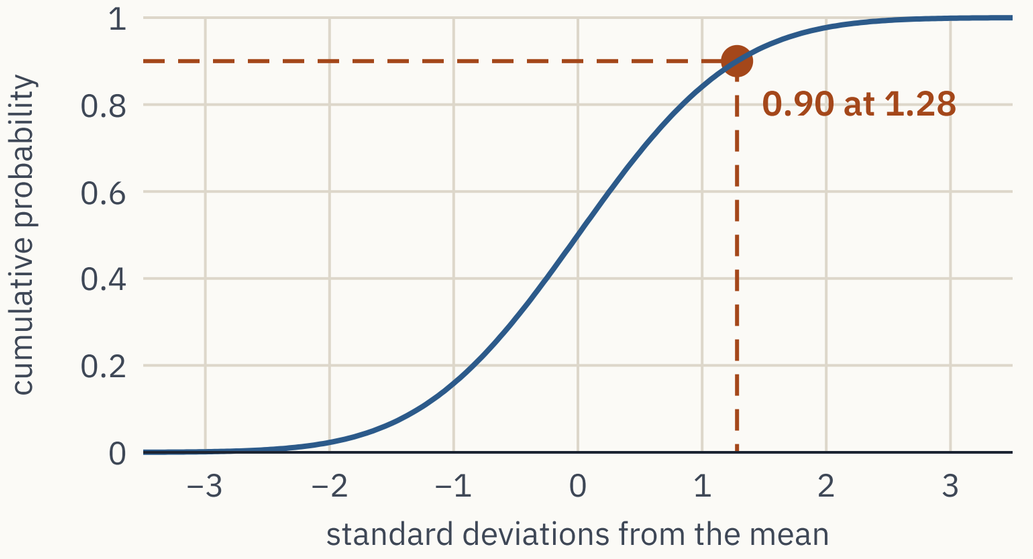
BEFORE THE DATA: WHAT A 90TH PERCENTILE IS

90% below it, 10% above

A normal distribution



Its cumulative distribution function



Lead results are not normal, so the rule finds this point by rank.

LIVE: THE RULE ON TEN VALUES

Sort, multiply by 0.9, read that rank

The first ten samples' first draws, sorted, in µg/L

Rank	1	2	3	4	5	6	7	8	9	10
µg/L	0.344	1.1	1.111	1.951	6.2	7.2	8.007	8.133	10.6	40.63

The rule: 10.6 µg/L

0.9 × 10 = 9, so the value at rank 9. Python counts from 0, so rank 9 is index 8.

np.percentile: 13.603

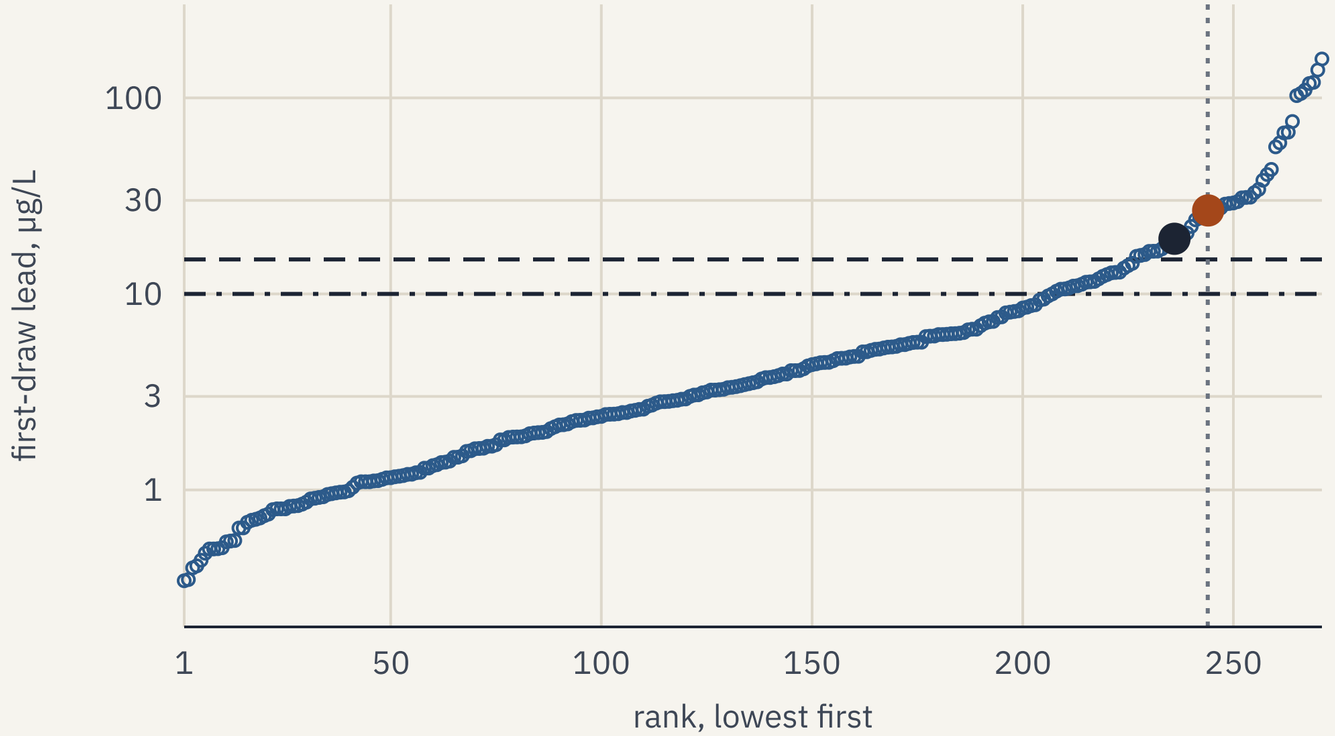
Interpolated a tenth of the way to rank 10: a value no home measured.

Index 9: 40.63

np.sort(x)[int(0.9 * 10)] counts from 0 and lands on rank 10, the largest.

SECTION 1: ALL 271 FIRST DRAWS

The mean passes; the rule does not



The lead axis is logarithmic.

15 µg/L, 2015

10 µg/L, from 2027

Rank 244, 26.64 µg/L

Rank 236, 19.12 µg/L

Mean 10.65 µg/L, below 15. But 45 of 271 samples are above 15, more than one in ten.

0.9 × 271 = 243.9 (dotted line). The 90th percentile, 26.64 µg/L, is an exceedance under the rule. The lower bound at rank 236 shows it at 95%.

ASSIGNMENTS

One draw of 60 samples each

#	Draw
1	draw_01
2	draw_02
3	draw_03
4	draw_04
5	draw_05
6	draw_06
7	draw_07

#	Draw
8	draw_08
9	draw_09
10	draw_10
11	draw_11
12	draw_12
13	draw_13
14	draw_14

#	Draw
15	draw_15
16	draw_16
17	draw_17
18	draw_18
19	draw_19
20	draw_20

#	Draw
21	draw_21
22	draw_22
23	draw_23
24	draw_24
25	draw_25
26	draw_26

In Parameters, set **DRAW** to your draw, such as DRAW = "draw_01", run the Parameters cell, then the cell after TASK 1. Section 1 is the worked example; section 2 is your 60 samples.

AI PRACTICE 3 OF 8, AGAIN: GIVE THE ASSISTANT A CHECK TO RUN

Write the check before you ask

Live, on the ten values

1. First, in a markdown cell:

Check: the sorted values with their ranks. Accept if rank 9 matches the assistant’s number and differs from np.percentile.

2. Then ask for code that sets assistant_p90 and prints that check.

3. Paste it into a new cell, run it, read rank 9: 10.6 µg/L.

What makes a check

- It runs in your notebook, under a cell, not in the chat.
- It prints the values the answer came from, not only the answer.
- Each wrong method you can name would print something different.

The sentence in the chat is a claim. The printout is the check. A check written before the answer cannot be bent to fit it.

TASK 3 AND TASK 4, AFTER YOUR 90TH PERCENTILE

Your check, your prompt, your cells

TASK 3, after you save the notebook

1. Add the prompt record and fill in its second line: your check, and what you will accept.
2. Write your own prompt for assistant_p90 and code that prints your check. Code in the chat.
3. Add a code cell, paste all of the code into it, run it, keep it.
4. Read the printout, then fill in the other two lines.

The record, a markdown cell

```
**Prompt record**
- **My prompt:**
- **The check I asked for, and what I will accept:**
- **What the printout showed:**
```

TASK 4: a code cell you write that sets rank_bound and bound_ug_L, as section 1's last cell did for all 271.

The record and a code cell that runs both count toward execution. You compare checks with a neighbor after TASK 5.

THE JUDGMENT CELL, SECTION 4

Your call on the city, and one entry

(a) Your answer

- Is your draw an exceedance of 15 and of 10 µg/L under the rule, and does its lower bound show it at 95%?
- Your 90th percentile at rank 54, its bound, and the first-draw minus flushed difference with the interval from the test you chose
- What the flushed bottle would have called, and what your call means for residents

(b) One judgment entry

- What the assistant suggested
- What you did: changed, rejected, or verified
- Why, from a value your notebook printed

The rule makes the call: a 90th percentile above 15 is an exceedance. The bound says how strongly 60 homes show it.

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Fill in your prompt record, then Restart, Run All, save, and upload activity.ipynb by **1:30 pm Tuesday, November 3**.

The graders' run downloads the same file and checks it byte for byte.

Scoring, 0 to 3

1 if it runs after a restart and holds your prompt record, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.